The Theory of Rational Return and Risk

An Algebraic, Financial-Economic, Econometric and Multidisciplinary Framework

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Abstract

This treatise develops the Theory of Rational Return and Risk, a proposed mathematical framework for representing the relationship between financial return and risk through fractional-linear transformations.

The fundamental relationship is expressed as

$$\frac{ar+b}{cr+d} = \frac{fR+g}{pR+q}, \tag{1}$$

where r denotes return, R denotes risk, and the remaining coefficients are structural parameters.

Cross-multiplication establishes a bilinear relationship between return and risk. The resulting functional representation belongs to the family of Mobius transformations.

The treatise develops the algebraic foundations, comparative statics, investor optimization, portfolio equilibrium, rational asset pricing, stochastic discount factors, bond pricing, duration, convexity, nonlinear econometric estimation, instrumental variables, Generalized Method of Moments, Bayesian estimation, dynamic models and regime-switching extensions.

It further investigates potential applications to financial planning and analysis, welfare economics, institutional economics, political economy, international relations, sovereign finance, geopolitical risk, operational resilience, cybersecurity, biological systems, physical systems and computational intelligence.

Particular attention is given to the distinction between mathematical identities, economic assumptions and empirically testable propositions.

The framework is presented as a research programme. Its empirical superiority over established financial models remains an open question requiring systematic validation.

The treatise ends with "The End"

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1 Introduction

1.1 Motivation

Return and risk are two fundamental variables in financial economics.

Investment decisions require the evaluation of expected returns relative to uncertainty, downside exposure, liquidity constraints and the possibility of financial loss.

Classical portfolio theory establishes a relationship between expected return and portfolio variance [15].

The Capital Asset Pricing Model subsequently provides an equilibrium relationship between expected return and systematic risk [21, 13, 16].

Arbitrage pricing theory and multifactor asset-pricing models extend these relationships [20, 9].

The present treatise investigates a different mathematical representation.

We propose that return and risk can be connected through a fractional-linear relationship.

The central equation is

$$\boxed{\frac{ar+b}{cr+d} = \frac{fR+g}{pR+q}} \tag{2}$$

The principal objective is to determine the mathematical, economic and econometric implications of this structure.

1.2 Research objectives

The research has the following objectives:

1. Establish the algebraic foundations of the return-risk relationship.
2. Derive the reciprocal functions of return and risk.
3. Investigate monotonicity, curvature, asymptotes and singularities.
4. Develop investor optimization and equilibrium conditions.
5. Derive rational asset-pricing and bond-pricing relationships.
6. Establish an econometric framework for identification and estimation.
7. Develop dynamic and regime-dependent extensions.
8. Investigate applications to financial planning, public finance and systemic risk.
9. Develop computational algorithms for empirical validation.
10. Establish a multidisciplinary research programme.

1.3 Methodological distinction

Three categories of statements will be distinguished.

Mathematical results. Statements derived directly from algebraic assumptions.

Economic propositions. Statements that require additional assumptions about preferences, markets, institutions and equilibrium.

Empirical hypotheses. Statements that must be evaluated using observations and statistical inference.

This distinction prevents mathematical identities from being interpreted as empirically established economic laws.

2 Algebraic Foundations

2.1 The fundamental equation

We begin with

$$\frac{ar+b}{cr+d} = \frac{fR+g}{pR+q}. \tag{3}$$

The variables are:

$$r = \text{Rational Return}, \tag{4}$$
$$R = \text{Risk}. \tag{5}$$

The structural coefficients are

$$a, b, c, d, f, g, p, q \in \mathbb{R}. \tag{6}$$

Initially, we assume that all denominators are nonzero.
Cross-multiplication gives

$$(ar+b)(pR+q) = (cr+d)(fR+g). \tag{7}$$

Expanding:

$$aprR + aqr + bpR + bq = cfrR + cgr + dfR + dg. \tag{8}$$

Rearranging:

$$(ap-cf)rR + (aq-cg)r + (bp-df)R + (bq-dg) = 0. \tag{9}$$

Define the composite coefficients:

$$A = ap - cf, \tag{10}$$
$$B = aq - cg, \tag{11}$$
$$C = bp - df, \tag{12}$$
$$D = bq - dg. \tag{13}$$

The fundamental equation becomes

$$\boxed{ArR + Br + CR + D = 0.} \tag{14}$$

This is the fundamental bilinear representation of the theory.

2.2 Solving for return

Collecting the terms containing r:

$$r(AR + B) = -(CR + D). \tag{15}$$

Therefore,

$$\boxed{r(R) = -\frac{CR + D}{AR + B}.} \tag{16}$$

2.3 Solving for risk

Collecting the terms containing R:

$$R(Ar + C) = -(Br + D). \tag{17}$$

Therefore,

$$\boxed{R(r) = -\frac{Br + D}{Ar + C}.} \tag{18}$$

These two equations constitute the reciprocal functional representations.

2.4 Mobius transformation

A Mobius transformation has the general form

$$f(x) = \frac{\alpha x + \beta}{\gamma x + \delta}. \tag{19}$$

The rational return function is a Mobius transformation.
Its matrix representation is

$$\mathcal{M} = \begin{pmatrix} -C & -D \\ A & B \end{pmatrix}. \tag{20}$$

The determinant is

$$\boxed{\Delta = AD - BC.} \tag{21}$$

Definition 2.1 (Non-degenerate rational transformation)**.** *The rational return-risk transformation is non-degenerate if*

$$AD - BC \neq 0. \tag{22}$$

Theorem 2.2 (Reciprocal invertibility)**.** *If*

$$AD - BC \neq 0, \tag{23}$$

the rational return function is invertible on its admissible domain.
Its inverse is

$$R(r) = -\frac{Br + D}{Ar + C}. \tag{24}$$

Proof. Starting with

$$r = -\frac{CR + D}{AR + B}, \tag{25}$$

we obtain

$$r(AR + B) = -CR - D. \tag{26}$$

Collecting terms:

$$R(Ar + C) = -(Br + D). \tag{27}$$

Therefore,

$$R = -\frac{Br + D}{Ar + C}. \tag{28}$$

The non-degeneracy condition ensures that the corresponding Mobius transformation is invertible as a transformation of the projective line.

□

2.5 Composition and projective representation

Mobius transformations possess a composition structure.

If two transformations are represented by matrices $\mathcal{M}_1$ and $\mathcal{M}_2$, their composition is represented by

$$\mathcal{M}_{21} = \mathcal{M}_2\mathcal{M}_1. \tag{29}$$

Multiplication of a Mobius matrix by a nonzero scalar does not change the represented function.

Consequently, the structural coefficients are identified only up to a common nonzero scale.

This observation will become important in the econometric sections.

3 Comparative Statics and Geometry

3.1 First derivative

Differentiating the return function:

$$r(R) = -\frac{CR + D}{AR + B}, \tag{30}$$

we obtain

$$\boxed{\frac{dr}{dR} = \frac{AD - BC}{(AR + B)^2}.} \tag{31}$$

Proposition 3.1 (Monotonicity)**.** *For a non-degenerate transformation, return is strictly monotonic in risk on every connected admissible interval.*

Proof. Since

$$(AR + B)^2 > 0 \tag{32}$$

wherever the function is defined, the sign of the derivative is determined entirely by

$$AD - BC. \tag{33}$$

Therefore the derivative has a constant nonzero sign on each connected admissible interval.

□

The possible cases are:

$$AD - BC > 0 \implies \frac{dr}{dR} > 0, \tag{34}$$

$$AD - BC < 0 \implies \frac{dr}{dR} < 0. \tag{35}$$

3.2 Second derivative

Differentiating again:

$$\boxed{\frac{d^2r}{dR^2} = -\frac{2A(AD - BC)}{(AR + B)^3}.} \tag{36}$$

This expression determines curvature.

3.3 Inverse derivative

Similarly,

$$\boxed{\frac{dR}{dr} = \frac{AD - BC}{(Ar + C)^2}.} \tag{37}$$

At corresponding points:

$$\boxed{\frac{dr}{dR}\frac{dR}{dr} = 1.} \tag{38}$$

3.4 Asymptotic return

For $A \neq 0$,

$$\boxed{\lim_{R \to \pm\infty} r(R) = -\frac{C}{A}.} \tag{39}$$

The horizontal asymptote is therefore

$$r_\infty = -\frac{C}{A}. \tag{40}$$

3.5 Critical risk and singularity

The denominator vanishes at

$$\boxed{R_c = -\frac{B}{A}.} \tag{41}$$

If the numerator does not simultaneously vanish, this point is a pole.
The economically admissible risk domain must exclude this singularity.
For example, a model may impose

$$R \geq 0, \tag{42}$$
$$AR + B > 0. \tag{43}$$

These are economic domain restrictions rather than universal algebraic necessities.

3.6 Geometrical interpretation

The bilinear equation

$$ArR + Br + CR + D = 0 \tag{44}$$

defines a curve in the return-risk plane.
The curve is generally a rectangular hyperbola after appropriate affine transformations.

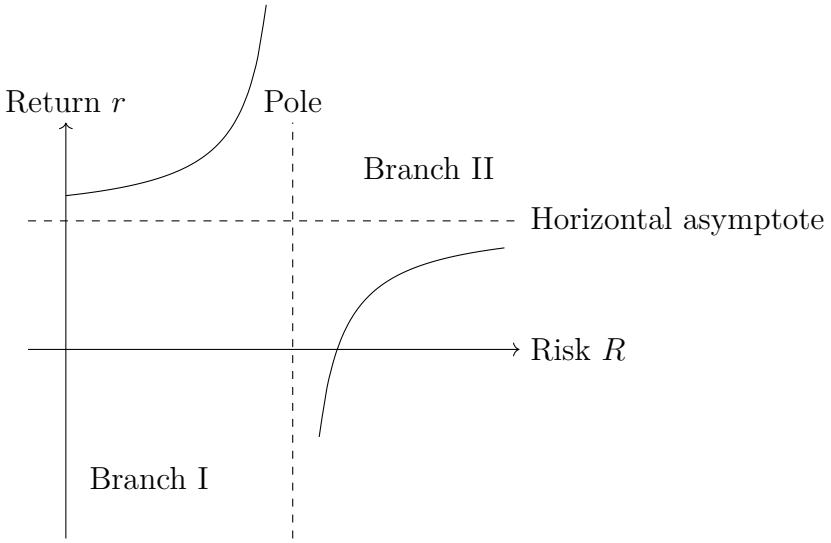


Figure 1: Schematic geometry of a rational return-risk transformation.

The figure is illustrative and does not represent a particular estimated parameter configuration.

4 Financial-Economic Foundations

4.1 Economic interpretation

We interpret r as expected investment return and R as a specified measure of financial risk.

Possible definitions of risk include:

Variance. The expected squared deviation of returns from their conditional mean.

Volatility. The square root of conditional variance.

Downside deviation. A measure of losses below a specified threshold.

Value at Risk. A quantile-based loss measure.

Expected Shortfall. The expected loss conditional on exceeding a specified VaR threshold.

Default probability. The probability of contractual non-performance.

These measures are not interchangeable.

The interpretation of the coefficients depends on the selected risk measure and its units.

4.2 Risk premium

Let r_f denote the risk-free rate.

Define the risk premium:

$$\pi(R) = r(R) - r_f. \tag{45}$$

Substitution gives

$$\boxed{\pi(R) = -\frac{(C + Ar_f)R + (D + Br_f)}{AR + B}.} \tag{46}$$

The marginal risk premium is

$$\boxed{\frac{d\pi}{dR} = \frac{AD - BC}{(AR + B)^2}.} \tag{47}$$

4.3 Zero-risk normalization

If the model is required to satisfy

$$r(0) = r_f, \tag{48}$$

then

$$-\frac{D}{B} = r_f. \tag{49}$$

For $B \neq 0$:

$$\boxed{D = -Br_f.} \tag{50}$$

The risk premium becomes

$$\boxed{\pi(R) = -\frac{(C + Ar_f)R}{AR + B}.} \tag{51}$$

This is a useful normalization for financial applications.

It should not be imposed when the selected risk measure does not have a meaningful zero-risk state.

4.4 Investor preferences

Classical expected utility theory provides a foundation for investor decision-making [2, 6].

We introduce the proposed rational utility function:

$$\boxed{U_R(r, R) = r - \lambda R + \eta r R.} \tag{52}$$

Here:

$$\lambda > 0, \tag{53}$$
$$\eta \in \mathbb{R}. \tag{54}$$

The marginal utilities are

$$\frac{\partial U_R}{\partial r} = 1 + \eta R, \tag{55}$$
$$\frac{\partial U_R}{\partial R} = -\lambda + \eta r. \tag{56}$$

The interaction term permits the marginal utility of return to depend on risk.

For a rational investor, a suitable admissible domain may require

$$1 + \eta R > 0. \tag{57}$$

This ensures positive marginal utility of return.

4.5 Investor optimization

The investor solves

$$\boxed{\max_{R \in \mathcal{R}} U_R(r(R), R).} \tag{58}$$

The first-order condition for an interior optimum is

$$\boxed{(1 + \eta R)r'(R) + \eta r(R) - \lambda = 0.} \tag{59}$$

Substituting the Mobius derivatives:

$$\boxed{(1+\eta R)\frac{AD-BC}{(AR+B)^2}-\eta\frac{CR+D}{AR+B}-\lambda=0.} \tag{60}$$

The second-order condition for a strict local maximum is

$$\boxed{(1+\eta R)r''(R)+2\eta r'(R)<0.} \tag{61}$$

Theorem 4.1 (Existence of rational equilibrium)**.** *Suppose the admissible risk set is compact and nonempty, and the rational utility function is continuous on that set.*

Then an optimal risk choice exists.

If the optimum is interior and the utility function is differentiable, it satisfies the first-order condition.

Proof. The existence of an optimum follows from the extreme value theorem.

At an interior optimum, differentiability requires the derivative of utility with respect to risk to equal zero.

Applying the chain rule gives the stated condition.

□

4.6 Special case: additive risk aversion

Setting

$$\eta=0 \tag{62}$$

gives

$$U_R(r,R)=r-\lambda R. \tag{63}$$

The first-order condition becomes

$$\frac{AD-BC}{(AR+B)^2}=\lambda. \tag{64}$$

For $A\neq 0$, the stationary candidates are

$$\boxed{R^*=\frac{\pm\sqrt{(AD-BC)/\lambda}-B}{A}.} \tag{65}$$

Only candidates satisfying the admissible domain and second-order condition are economically valid.

4.7 Portfolio optimization

Let the investor hold n assets.

The portfolio weights are represented by

$$\mathbf{w}=(w_1,w_2,\ldots,w_n)^\top. \tag{66}$$

Expected returns are represented by

$$\boldsymbol{\mu}=(\mu_1,\mu_2,\ldots,\mu_n)^\top. \tag{67}$$

Let $\mathbf{\Sigma}$ denote the covariance matrix.
Portfolio return is

$$r_p = \mathbf{w}^\top \boldsymbol{\mu}. \tag{68}$$

Portfolio variance is

$$R_p = \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}. \tag{69}$$

A quadratic benchmark is

$$U(\mathbf{w}) = \mathbf{w}^\top \boldsymbol{\mu} - \lambda \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}. \tag{70}$$

The unconstrained first-order condition is

$$\boldsymbol{\mu} - 2\lambda \mathbf{\Sigma} \mathbf{w} = 0. \tag{71}$$

Therefore,

$$\boxed{\mathbf{w}^* = \frac{1}{2\lambda} \mathbf{\Sigma}^{-1} \boldsymbol{\mu}.} \tag{72}$$

This is the classical quadratic-utility benchmark [15].
The rational return-risk extension is

$$\boxed{\mathbf{w}^\top \boldsymbol{\mu} = -\frac{C \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w} + D}{A \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w} + B}.} \tag{73}$$

This imposes a common variance-return schedule on the portfolios being considered. It is a restrictive hypothesis and should be tested rather than assumed universally.

5 General Equilibrium and Welfare Economics

5.1 Heterogeneous investors

Let i index investors.
Each investor has wealth W_i, risk aversion λ_i, and portfolio weights $\mathbf{w}_i$.
The optimization problem is

$$\max_{\mathbf{w}_i} U_i(r_i, R_i). \tag{74}$$

Market clearing for asset j requires

$$\boxed{\sum_{i=1}^{N} W_i w_{ij} = S_j,} \tag{75}$$

where S_j is the outstanding supply.
This condition connects individual optimization to market equilibrium.

5.2 Welfare economics

Social welfare can be represented by

$$\mathcal{W} = \sum_{i=1}^{N} \omega_i U_i(r_i, R_i), \tag{76}$$

where ω_i are social welfare weights.

The social planner solves

$$\max_{\{\mathbf{w}_i\}} \mathcal{W} \tag{77}$$

subject to resource and market-clearing constraints.

The welfare implications of the rational return-risk relationship depend on the distribution of wealth, risk exposure, externalities and institutional arrangements.

The mathematical return-risk function alone does not determine social welfare.

5.3 Public finance

Governments face financial trade-offs between borrowing costs and fiscal risk.

Let:

$$r_g = \text{Sovereign borrowing rate}, \tag{78}$$
$$R_g = \text{Sovereign financial risk}. \tag{79}$$

A proposed sovereign application is

$$r_g = -\frac{C_g R_g + D_g}{A_g R_g + B_g}. \tag{80}$$

The specification can be evaluated against sovereign yields, credit ratings, debt ratios, foreign currency exposure and fiscal indicators.

6 Rational Asset Pricing

6.1 One-period asset pricing

Let X_{t+1} denote a future payoff.

The expected return is

$$r_t = \frac{\mathbb{E}_t[X_{t+1}] - P_t}{P_t}. \tag{81}$$

Rearranging:

$$P_t = \frac{\mathbb{E}_t[X_{t+1}]}{1 + r_t}. \tag{82}$$

Using the rational return-risk function:

$$r_t = -\frac{CR_t + D}{AR_t + B}, \tag{83}$$

we obtain

$$\boxed{P_t = \frac{\mathbb{E}_t[X_{t+1}](AR_t + B)}{(A - C)R_t + (B - D)}.} \tag{84}$$

The admissible pricing domain requires a nonzero denominator and a positive gross return.

6.2 Price sensitivity

Assume the expected payoff is independent of risk.

Differentiating equation (84):

$$\boxed{\frac{dP_t}{dR_t} = -\frac{\mathbb{E}_t[X_{t+1}](AD - BC)}{[(A - C)R_t + (B - D)]^2}.} \tag{85}$$

The negative sign is important.

If the determinant is positive, the required return increases with risk, and the price of a fixed positive expected payoff decreases with risk.

This is consistent with conventional discounted-cash-flow reasoning.

6.3 Stochastic discount factor

Modern asset pricing can be expressed through the stochastic discount factor [14, 5, 7].

Let M_{t+1} denote the stochastic discount factor.

The fundamental pricing equation is

$$\boxed{P_t = \mathbb{E}_t[M_{t+1}X_{t+1}].} \tag{86}$$

For gross return

$$G_{t+1} = \frac{X_{t+1}}{P_t}, \tag{87}$$

the pricing restriction is

$$\mathbb{E}_t[M_{t+1}G_{t+1}] = 1. \tag{88}$$

Using the covariance identity:

$$\mathbb{E}_t[MG] = \mathbb{E}_t[M]\mathbb{E}_t[G] + \mathrm{Cov}_t(M, G), \tag{89}$$

we obtain

$$\boxed{\mathbb{E}_t[G] = \frac{1 - \mathrm{Cov}_t(M, G)}{\mathbb{E}_t[M]}.} \tag{90}$$

The risk-free gross return is

$$G_{f,t} = \frac{1}{\mathbb{E}_t[M]}. \tag{91}$$

Therefore,

$$\boxed{\mathbb{E}_t[G] - G_{f,t} = -\frac{\mathrm{Cov}_t(M,G)}{\mathbb{E}_t[M]}.} \tag{92}$$

This provides a theoretical connection between risk compensation and the pricing kernel.

An empirical equivalence between the rational risk measure and pricing-kernel covariance must be established separately.

7 Rational Bond Pricing

7.1 Zero-coupon bonds

Consider a zero-coupon bond paying one monetary unit at maturity T.

Let y_T be its annual effective yield.

Then

$$P(0,T) = (1+y_T)^{-T}. \tag{93}$$

If

$$y_T = -\frac{CR_T + D}{AR_T + B}, \tag{94}$$

then

$$\boxed{P(0,T) = \left[\frac{AR_T + B}{(A-C)R_T + (B-D)}\right]^T.} \tag{95}$$

7.2 Coupon-bearing bonds

For cash flows CF_k:

$$P_0 = \sum_{k=1}^{T} \frac{CF_k}{(1+y_k)^k}. \tag{96}$$

The rational risk-adjusted representation is

$$\boxed{P_0 = \sum_{k=1}^{T} CF_k \left[\frac{AR_k + B}{(A-C)R_k + (B-D)}\right]^k.} \tag{97}$$

This formulation allows the risk measure to vary across maturities.

7.3 Duration

For a zero-coupon bond:

$$P = (1+y)^{-T}. \tag{98}$$

The first derivative is

$$\frac{dP}{dy} = -T(1+y)^{-T-1}. \tag{99}$$

The second derivative is

$$\frac{d^2P}{dy^2} = T(T+1)(1+y)^{-T-2}. \tag{100}$$

Using the rational return-risk relationship:

$$\boxed{\frac{dP}{dR} = -T(1+r)^{-T-1}\frac{AD-BC}{(AR+B)^2}.} \tag{101}$$

7.4 Risk convexity

By the chain rule:

$$\frac{d^2P}{dR^2} = \frac{d^2P}{dy^2}\left(\frac{dy}{dR}\right)^2 + \frac{dP}{dy}\frac{d^2y}{dR^2}. \tag{102}$$

Therefore,

$$\boxed{\begin{aligned}\frac{d^2P}{dR^2} &= T(T+1)(1+r)^{-T-2}\left(\frac{AD-BC}{(AR+B)^2}\right)^2 \\ &\quad + 2TA(1+r)^{-T-1}\frac{AD-BC}{(AR+B)^3}.\end{aligned}} \tag{103}$$

This expression represents the second-order sensitivity of bond prices to the chosen risk measure.

It is distinct from conventional yield convexity, although it is mathematically related to it.

8 Theoretical Econometrics

8.1 Structural econometric equation

We introduce stochastic disturbances:

$$r_t = -\frac{CR_t + D}{AR_t + B} + \epsilon_t. \tag{104}$$

This is the nonlinear structural return equation.

An equivalent implicit representation is

$$Ar_tR_t + Br_t + CR_t + D = u_t. \tag{105}$$

The two error specifications are not identical: the disturbance in the implicit equation is scaled differently from an additive disturbance in the explicit return equation.

This distinction must be respected in estimation.

8.2 Scale identification

The Mobius transformation is unchanged under

$$(A, B, C, D) \longrightarrow (kA, kB, kC, kD), \tag{106}$$

for any nonzero scalar k.

Consequently, the four coefficients cannot be identified independently without normalization.

If $D \neq 0$, one possible normalization is

$$D = 1. \tag{107}$$

Then

$$\theta_1 = \frac{A}{D}, \qquad \theta_2 = \frac{B}{D}, \qquad \theta_3 = \frac{C}{D}. \tag{108}$$

If $D = 0$, another nonzero coefficient must be normalized.

8.3 Nonlinear least squares

The parameter vector is

$$\boldsymbol{\beta} = (A, B, C, D)^\top. \tag{109}$$

The nonlinear least squares estimator is

$$\boxed{\widehat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \sum_{t=1}^{T} \left[r_t + \frac{CR_t + D}{AR_t + B} \right]^2.} \tag{110}$$

This requires an appropriate normalization and admissible denominator restrictions.

8.4 Jacobian and local identification

Define the residual:

$$e_t = r_t + \frac{CR_t + D}{AR_t + B}. \tag{111}$$

The gradient is

$$\frac{\partial e_t}{\partial A} = -\frac{R_t(CR_t + D)}{(AR_t + B)^2}, \tag{112}$$

$$\frac{\partial e_t}{\partial B} = -\frac{CR_t + D}{(AR_t + B)^2}, \tag{113}$$

$$\frac{\partial e_t}{\partial C} = \frac{R_t}{AR_t + B}, \tag{114}$$

$$\frac{\partial e_t}{\partial D} = \frac{1}{AR_t + B}. \tag{115}$$

Let J denote the Jacobian matrix.

Local identification requires the appropriate rank condition after normalization.

The determinant of the transformation must also be evaluated separately.

8.5 Endogeneity

Return and risk may be jointly determined.

We introduce a risk-generation equation:

$$R_t = \gamma_0 + \gamma_1 r_t + \boldsymbol{\gamma}_2^\top \mathbf{Z}_t + \eta_t. \tag{116}$$

The return equation is

$$r_t = -\frac{CR_t + D}{AR_t + B} + \epsilon_t. \tag{117}$$

If

$$\mathrm{Cov}(\epsilon_t, \eta_t) \neq 0, \tag{118}$$

the endogenous relationship must be treated explicitly.

8.6 Instrumental variables

Let H_t be an instrument vector.

The conditions are

$$\mathbb{E}[H_t \epsilon_t] = 0, \tag{119}$$

$$\mathrm{Cov}(H_t, R_t) \neq 0. \tag{120}$$

The first condition represents instrument exogeneity.

The second represents relevance.

The exclusion restriction must be justified economically and institutionally.

8.7 Generalized Method of Moments

Define

$$e_t(\boldsymbol{\beta}) = r_t + \frac{CR_t + D}{AR_t + B}. \tag{121}$$

The moment condition is

$$\boxed{\mathbb{E}[H_t e_t(\boldsymbol{\beta})] = 0.} \tag{122}$$

The sample moment vector is

$$g_T(\boldsymbol{\beta}) = \frac{1}{T} \sum_{t=1}^{T} H_t e_t(\boldsymbol{\beta}). \tag{123}$$

The GMM estimator is

$$\boxed{\widehat{\boldsymbol{\beta}}_{GMM} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}}\ g_T(\boldsymbol{\beta})^\top W_T g_T(\boldsymbol{\beta}).} \tag{124}$$

Hansen's GMM framework provides the econometric foundation [11].

For financial time series, the weighting matrix should account for heteroskedasticity and serial dependence [22, 17].

8.8 Simultaneous nonlinear GMM

Define the structural disturbances:

$$\epsilon_t = r_t + \frac{CR_t + D}{AR_t + B}, \tag{125}$$

$$\eta_t = R_t - \gamma_0 - \gamma_1 r_t - \boldsymbol{\gamma}_2^\top \mathbf{Z}_t. \tag{126}$$

The combined moment vector is

$$\mathbf{m}_t = \begin{pmatrix} H_t \epsilon_t \\ Z_t \eta_t \end{pmatrix}. \tag{127}$$

The system estimator minimizes

$$\boxed{Q_T(\boldsymbol{\theta}) = g_T(\boldsymbol{\theta})^\top W_T g_T(\boldsymbol{\theta}).} \tag{128}$$

The required rank and exclusion restrictions must be established for the complete system.

9 Bayesian Econometrics

9.1 Bayesian formulation

Let $\boldsymbol{\theta}$ represent the normalized structural parameters.

Bayes' theorem gives

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathcal{D})}. \tag{129}$$

Here:

$$p(\boldsymbol{\theta}) = \text{Prior distribution}, \tag{130}$$

$$p(\mathcal{D} \mid \boldsymbol{\theta}) = \text{Likelihood}, \tag{131}$$

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \text{Posterior distribution}. \tag{132}$$

The likelihood requires explicit assumptions about the distribution of structural errors.

9.2 Posterior inference

Relevant posterior statistics include:

$$\mathbb{E}[\boldsymbol{\theta} \mid \mathcal{D}] = \text{Posterior mean}, \tag{133}$$

$$\text{Var}(\boldsymbol{\theta} \mid \mathcal{D}) = \text{Posterior covariance}. \tag{134}$$

We can also evaluate:

$$\Pr(AD - BC > 0 \mid \mathcal{D}). \tag{135}$$

This measures posterior support for a positive marginal return-risk relationship.

9.3 Monte Carlo estimation

A generic Metropolis-Hastings procedure is:

1. Specify a normalized parameter vector.
2. Specify prior distributions.
3. Construct the likelihood.
4. Draw a candidate parameter vector.
5. Evaluate the posterior density.
6. Accept or reject the candidate according to the Metropolis-Hastings probability.
7. Repeat until convergence diagnostics are satisfactory.
8. Calculate posterior summaries.

10 Dynamic Rational Return-Risk Models

10.1 Time-varying coefficients

The coefficients may vary over time:

$$r_t = -\frac{C_t R_t + D_t}{A_t R_t + B_t} + \epsilon_t. \tag{136}$$

Define the latent state:

$$\mathbf{s}_t = (A_t, B_t, C_t, D_t)^\top. \tag{137}$$

A random-walk state equation is

$$\mathbf{s}_t = \mathbf{s}_{t-1} + \boldsymbol{\nu}_t. \tag{138}$$

The observation equation is nonlinear.
Possible estimation techniques include:

Extended Kalman Filter. Uses a local linear approximation.

Unscented Kalman Filter. Uses deterministic sigma-point transformations.

Particle Filter. Uses sequential simulation of latent states.

Bayesian Filtering. Combines prior state distributions with new observations.

A consistent normalization is required at every time step.

10.2 Conditional volatility

A possible extension uses a GARCH risk process [8, 4].

Let h_t denote conditional variance.

A GARCH(1,1) process is

$$h_t = \omega + \alpha \epsilon_{t-1}^2 + \beta h_{t-1}. \tag{139}$$

We may define

$$R_t = \sqrt{h_t}. \tag{140}$$

The rational return equation becomes

$$r_t = -\frac{C\sqrt{h_t} + D}{A\sqrt{h_t} + B} + \epsilon_t. \tag{141}$$

This creates a nonlinear conditional-return model linked to conditional volatility.

10.3 Regime switching

Let the financial regime be

$$S_t \in \{1, 2, \ldots, K\}. \tag{142}$$

The regime-dependent equation is

$$\boxed{r_t = -\frac{C_{S_t} R_t + D_{S_t}}{A_{S_t} R_t + B_{S_t}} + \epsilon_t.} \tag{143}$$

Transition probabilities are

$$P_{ij} = \Pr(S_t = j \mid S_{t-1} = i). \tag{144}$$

The Markov-switching framework follows Hamilton's approach [10].

Potential regimes include:

Regime 1. Monetary stability.

Regime 2. Financial expansion.

Regime 3. Financial stress.

Regime 4. Systemic crisis.

These categories are proposed classifications and require empirical validation.

11 Financial Planning and Analysis

11.1 Capital budgeting

Let CF_t represent project cash flows.

The net present value is

$$NPV = \sum_{t=1}^{T} \frac{CF_t}{(1+r_t)^t} - I_0. \tag{145}$$

Using rational return:

$$NPV = \sum_{t=1}^{T} CF_t \left[\frac{AR_t + B}{(A-C)R_t + (B-D)} \right]^t - I_0. \tag{146}$$

This is a possible risk-adjusted capital budgeting application.

The discount rate must be consistent with the currency, inflation basis, cash-flow definition and risk treatment.

11.2 Scenario analysis

Let:

$$R_t^{(s)} \tag{147}$$

denote risk under scenario s.

The corresponding return is

$$r_t^{(s)} = -\frac{CR_t^{(s)} + D}{AR_t^{(s)} + B}. \tag{148}$$

A financial planning system can compare:

Baseline scenario. Expected economic conditions.

Adverse scenario. Higher volatility, weaker liquidity and lower asset values.

Severe scenario. Systemic crisis and market disruption.

Recovery scenario. Normalization of risk and financing conditions.

11.3 Corporate financial analysis

The framework can be applied to:

- Weighted average cost of capital.
- Capital structure.
- Debt-service capacity.
- Liquidity risk.

- Investment hurdle rates.
- Corporate valuation.
- Financial stress testing.

However, the risk variable must be defined consistently with the financial decision.

12 Statistics and Empirical Validation

12.1 Testable hypotheses

The principal hypotheses are summarized below.

Table 1: Principal theoretical hypotheses

Test	Restriction	Interpretation
H1	$AD - BC > 0$	Positive return-risk slope
H2	$AD - BC < 0$	Negative return-risk slope
H3	$D + Br_f = 0$	Zero-risk normalization
H4	$A = 0$	Affine special case
H5	Constant coefficients	Structural stability
H6	Variable coefficients	Structural change

12.2 Structural break testing

The stability hypothesis is

$$H_0 : \boldsymbol{\theta}_1 = \boldsymbol{\theta}_2 = \cdots = \boldsymbol{\theta}_T. \tag{149}$$

The alternative allows coefficient changes.
Potential procedures include:

- Wald tests.
- Likelihood-ratio tests.
- Supremum Wald tests.
- Bai-Perron multiple-break procedures.
- Bayesian change-point analysis.

See Bai and Perron [3].

12.3 Forecast evaluation

Given a forecast risk level:

$$\widehat{R}_{t+1}, \tag{150}$$

the return forecast is

$$\boxed{\widehat{r}_{t+1} = -\frac{\widehat{C}\widehat{R}_{t+1} + \widehat{D}}{\widehat{A}\widehat{R}_{t+1} + \widehat{B}}.} \tag{151}$$

Forecast evaluation should include:

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_t - \widehat{r}_t)^2}, \tag{152}$$

$$MAE = \frac{1}{T}\sum_{t=1}^{T}|r_t - \widehat{r}_t|. \tag{153}$$

Out-of-sample forecasting is essential.
An in-sample fit does not establish predictive superiority.

13 Computer Science and Algorithms

13.1 Computational architecture

A computational implementation requires:

1. Data acquisition.
2. Data validation.
3. Return construction.
4. Risk measurement.
5. Structural parameter normalization.
6. Nonlinear estimation.
7. Identification diagnostics.
8. Forecast generation.
9. Stability testing.
10. Reproducibility validation.

13.2 Pseudocode

```
ALGORITHM: RATIONAL RETURN-RISK ESTIMATION

INPUT:
    Price series P
    Risk specification RISK
    Instrument set Z
    Initial parameter vector theta

PROCEDURE:

1. Validate timestamps and observations.

2. Calculate returns:
       r[t] = P[t] / P[t-1] - 1

3. Calculate risk:
       R[t] = RISK(r[t], historical_data)

4. Select parameter normalization.

5. Check admissible denominators:
       A*R[t] + B != 0

6. Define residual:
       e[t] = r[t] +
              (C*R[t] + D)/(A*R[t] + B)

7. Estimate parameters using NLS or GMM.

8. Evaluate:
       determinant = A*D - B*C

9. Calculate:
       marginal_return = determinant /
                         (A*R[t] + B)^2

10. Test structural stability.

11. Generate out-of-sample forecasts.

12. Compare against benchmark models.

OUTPUT:
    Parameter estimates
    Confidence intervals
    Determinant
```

```
Forecasts
Diagnostic statistics
Reproducibility report
```

13.3 Numerical safeguards

The implementation should monitor:

- Denominator proximity to zero.
- Overflow and underflow.
- Parameter scale indeterminacy.
- Ill-conditioned Jacobians.
- Numerical optimization convergence.
- Missing observations.
- Data leakage.
- Look-ahead bias.

13.4 Artificial intelligence and machine learning

Machine learning can be used to estimate time-varying risk or regime probabilities.

Potential architectures include:

Neural Networks. Estimate nonlinear mappings from market features to risk.

Transformers. Model temporal relationships across financial observations.

State-Space Models. Estimate latent financial conditions.

Reinforcement Learning. Explore sequential portfolio decisions.

Multi-Agent Learning. Model heterogeneous financial institutions.

A useful hybrid architecture is

$$R_t = f_\phi(\mathbf{X}_t), \tag{154}$$

followed by

$$r_t = -\frac{CR_t + D}{AR_t + B} + \epsilon_t. \tag{155}$$

The neural network estimates risk, while the rational transformation imposes a structured return-risk relationship.

This hybrid model must be evaluated against flexible black-box alternatives.

13.5 Cybersecurity

Financial data pipelines require controls for:

- Data integrity.
- Authentication.
- Access control.
- Encryption.
- Audit logging.
- Model versioning.
- Supply-chain security.
- Reproducible execution.

A corrupted price series can generate misleading return and risk estimates.
Cybersecurity is therefore part of the operational validity of the empirical framework.

14 Physics and Mathematical Systems

14.1 Fractional-linear transformations

Mobius transformations are important in complex analysis and geometry [1].
The transformation can be extended to complex-valued variables:

$$r(R) = -\frac{CR + D}{AR + B}, \qquad r, R \in \mathbb{C}. \tag{156}$$

The algebraic determinant remains

$$AD - BC. \tag{157}$$

This permits mathematical exploration of complex-valued interest rates and generalized financial fields.
Complex-valued financial variables require a separate economic interpretation.

14.2 Dynamical systems analogy

A time-varying return-risk system can be represented by

$$\frac{dr}{dt} = F(r, R, t). \tag{158}$$

If risk also evolves dynamically:

$$\frac{dR}{dt} = G(r, R, t), \tag{159}$$

the system becomes a coupled dynamical system.
Its equilibrium points satisfy

$$F(r^*, R^*, t) = 0, \tag{160}$$
$$G(r^*, R^*, t) = 0. \tag{161}$$

Local stability can be investigated using the Jacobian matrix of the dynamical system.

The analogy with physical dynamical systems is mathematical, not a claim that financial markets obey a universal physical law.

15 Chemistry and Biological Systems

15.1 Chemical kinetics

Chemical reaction rates depend on concentrations, temperature and reaction conditions.

A rational functional approximation may be written as

$$v(x) = \frac{\alpha x + \beta}{\gamma x + \delta}. \tag{162}$$

Such expressions can approximate saturation relationships over restricted domains.

The Michaelis-Menten equation provides a well-known biochemical example:

$$v(S) = \frac{V_{\max} S}{K_m + S}. \tag{163}$$

This is a special fractional-linear form.

The mathematical similarity does not imply that financial risk is chemically equivalent to substrate concentration.

15.2 Biological risk-return analogies

Biological systems frequently involve trade-offs between energy expenditure, survival probability and reproductive success.

A proposed abstract analogy is

$$B(x) = -\frac{\alpha x + \beta}{\gamma x + \delta}, \tag{164}$$

where x represents an exposure variable and $B(x)$ represents a biological response.

Such applications require independent biological foundations and empirical data.

16 Psychology, Psychiatry and Behaviour

16.1 Subjective risk

Objective risk and perceived risk should be distinguished.

Let:

$$R_t^o = \text{Objective risk}, \tag{165}$$
$$R_t^s = \text{Subjective risk perception}. \tag{166}$$

A possible perception model is

$$R_t^s = h(R_t^o, \mathcal{I}_t, \mathcal{B}_t), \tag{167}$$

where:

- $\mathcal{I}_t$ represents available information. - $\mathcal{B}_t$ represents beliefs and behavioural characteristics.

The rational return function may then use subjective risk:

$$r_t = -\frac{CR_t^s + D}{AR_t^s + B}. \tag{168}$$

Behavioural finance provides relevant foundations for examining deviations from classical rational-choice assumptions [12].

16.2 Behavioural asymmetry

Investors may react differently to gains and losses.

A possible extension is

$$r(R) = \begin{cases} -\dfrac{C_+R + D_+}{A_+R + B_+}, & R \in \mathcal{R}_+, \\ -\dfrac{C_-R + D_-}{A_-R + B_-}, & R \in \mathcal{R}_-. \end{cases} \tag{169}$$

This allows different parameter structures across behavioural regimes.

Psychiatric diagnoses should never be inferred from portfolio choices or risk-taking behaviour. Clinical interpretation requires independent medical evidence.

17 Law, Institutions and Governance

17.1 Contractual risk allocation

Financial contracts allocate risks between counterparties.

Examples include:

- Credit agreements.
- Derivative contracts.
- Insurance agreements.
- Sovereign debt contracts.
- Guarantees.
- Collateral arrangements.

A contract-specific return-risk function could be written as

$$r_c(R) = -\frac{C_cR + D_c}{A_cR + B_c}. \tag{170}$$

The coefficients would represent reduced-form contractual and institutional characteristics.

Legal enforceability, governing law, insolvency rules and collateral priority must be modelled separately.

The equation cannot replace legal analysis.

17.2 Institutional economics

Institutions influence transaction costs, information quality, enforcement and property rights.

Institutional economics provides a framework for examining these mechanisms [18, 19].

One possible extension is

$$r_t = -\frac{C(I_t)R_t + D(I_t)}{A(I_t)R_t + B(I_t)}, \tag{171}$$

where I_t represents institutional quality.

This is an empirical hypothesis requiring independent institutional measurement.

18 Political Economy, Geopolitics and Diplomacy

18.1 Sovereign risk

Sovereign financial risk can arise from:

- Public debt accumulation.
- Currency mismatches.
- Inflation instability.
- External financing dependence.
- Political instability.
- Institutional weakness.
- Financial sanctions.
- Geopolitical conflict.

A sovereign return-risk model is

$$r_g = -\frac{C_g R_g + D_g}{A_g R_g + B_g}. \tag{172}$$

The model could be estimated using sovereign bond yields and carefully constructed sovereign risk measures.

18.2 International financial relationships

Let i and j represent countries.

A bilateral financial relationship may be represented as

$$r_{ij} = -\frac{C_{ij}R_{ij} + D_{ij}}{A_{ij}R_{ij} + B_{ij}}. \tag{173}$$

Possible determinants include:

- Bilateral trade.
- Capital flows.
- Exchange-rate exposure.
- Monetary policy differences.
- Financial agreements.
- Sovereign credit conditions.
- Political and geopolitical uncertainty.

A network representation is

$$G = (V, E), \tag{174}$$

where vertices represent countries and edges represent financial relationships.

The rational return-risk transformation can be attached to individual network edges.

18.3 Diplomacy and international cooperation

Financial cooperation may reduce particular forms of risk through:

- Currency swap arrangements.
- Payment-system interoperability.
- Financial information sharing.
- Debt restructuring mechanisms.
- Trade agreements.
- Crisis coordination.

The effects of these institutions must be identified empirically rather than assumed.

19 Warfare Economics and National Resilience

19.1 Economic risk under conflict

Armed conflict can affect:

- Sovereign borrowing costs.
- Exchange rates.
- Commodity prices.
- Supply chains.
- Insurance premiums.
- Fiscal expenditure.
- Infrastructure resilience.

A conflict-dependent rational return-risk relationship may be represented as

$$r_t = -\frac{C_{S_t} R_t + D_{S_t}}{A_{S_t} R_t + B_{S_t}}, \tag{175}$$

where S_t represents the geopolitical or economic regime.

19.2 Defence economics

Let:

$$D_t = \text{Defence expenditure}, \tag{176}$$
$$Y_t = \text{National income}, \tag{177}$$
$$R_t = \text{Relevant national risk}. \tag{178}$$

A public-finance optimization problem may be

$$\max_{D_t} \left[W(Y_t, D_t) - \lambda R_t \right] \tag{179}$$

subject to fiscal and institutional constraints.

The theory can be used to examine financing trade-offs, sovereign debt sustainability and resource allocation.

It does not determine military strategy or operational targeting.

20 Linguistics and Information Theory

20.1 Semantic precision

The distinction between return, reward, risk, uncertainty and ambiguity is important.

These terms should not be treated as synonyms.

A linguistic representation of the theory requires consistent definitions across:

Return. A financial gain or loss relative to invested capital over a specified horizon.

Risk. A specified measure of uncertain financial outcomes.

Uncertainty. Incomplete knowledge of future outcomes.

Ambiguity. Uncertainty concerning the probability distribution itself.

Reward. A broader payoff concept that may include non-financial benefits.

20.2 Information quality

Let I_t denote the information available to market participants.

A possible extension is

$$R_t^s = h(R_t^o, I_t). \tag{180}$$

The distinction between objective and subjective risk becomes particularly important when information is incomplete or asymmetric.

21 Research Design and Empirical Programme

21.1 Proposed research stages

Table 2: Empirical research programme

Stage	Research activity	Principal output
1	Algebraic verification	Symbolic identities and proofs
2	Synthetic data	Parameter recovery and diagnostics
3	Asset-level estimation	Return-risk coefficient estimates
4	Portfolio analysis	Portfolio equilibrium evaluation
5	Fixed-income analysis	Bond pricing and convexity tests
6	Macroeconomic panel	Cross-country comparisons
7	Regime analysis	Structural break and crisis tests
8	Forecasting	Out-of-sample performance

21.2 Potential data sources

Relevant sources include:

- Reserve Bank of India.
- Federal Reserve Economic Data.
- Bank for International Settlements.
- International Monetary Fund.

- World Bank.
- OECD.
- National statistical authorities.
- National stock exchanges.
- Central securities depositories.
- Sovereign debt management offices.

Data must be harmonized across currencies, frequencies, accounting conventions and observation periods.

21.3 Benchmark models

The theory should be compared with:

Table 3: Benchmark financial models

Model	Main variable	Reference
Mean-Variance	Portfolio variance	Markowitz
CAPM	Systematic risk	Sharpe
APT	Multiple factors	Ross
Fama-French	Risk factors	Fama-French
GARCH	Conditional variance	Engle
SDF	Pricing kernel	Cochrane
Rational Model	Fractional-linear risk	Present theory

The proposed model should demonstrate incremental explanatory or predictive value.

22 Formal Limitations and Open Problems

22.1 Mathematical limitations

The Mobius representation imposes substantial structural restrictions.

In particular, the derivative has a constant sign on each connected admissible interval.

Consequently, the model cannot represent arbitrary non-monotonic return-risk relationships without additional regimes or functional extensions.

22.2 Economic limitations

The original equation does not independently derive:

- Investor preferences.
- Market-clearing conditions.
- Stochastic discount factors.

- Equilibrium risk premiums.
- Institutional constraints.

These require additional theoretical assumptions.

22.3 Econometric limitations

Important issues include:

- Scale normalization.
- Weak identification.
- Endogeneity.
- Measurement error.
- Nonstationarity.
- Structural breaks.
- Model misspecification.
- Denominator singularities.

22.4 Open mathematical problems

The following questions deserve further investigation:

1. Can the four composite coefficients be derived from primitive investor preferences?
2. Can the coefficients be derived from a stochastic discount factor?
3. Under what conditions does a general equilibrium generate a Mobius return-risk schedule?
4. Can the model be extended to multiple risk dimensions?
5. Can the theory accommodate ambiguity and non-probabilistic uncertainty?
6. Can a generalized cyclotomic or complex economic field representation be established?
7. Can the framework generate useful systemic early-warning indicators?

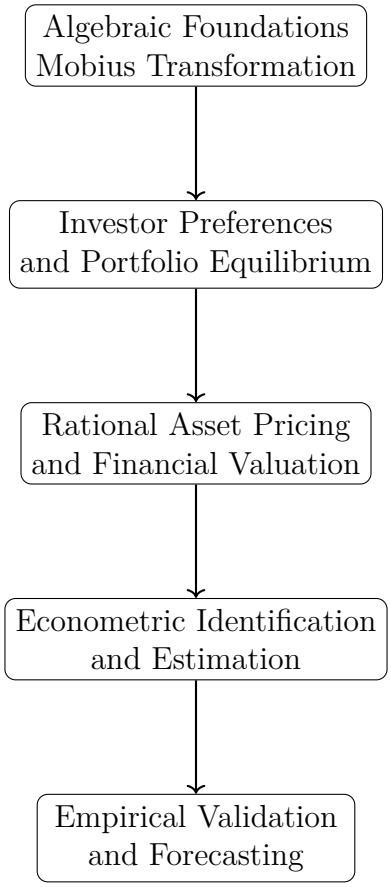


Figure 2: Unified architecture of the Theory of Rational Return and Risk.

23 Unified Theoretical Architecture

The proposed architecture consists of three connected systems.

The central structural equation is

$$\boxed{ArR + Br + CR + D = 0.} \tag{181}$$

Its functional representations are

$$\boxed{r(R) = -\frac{CR + D}{AR + B},} \tag{182}$$

$$\boxed{R(r) = -\frac{Br + D}{Ar + C}.} \tag{183}$$

The determinant

$$\boxed{\Delta = AD - BC} \tag{184}$$

governs invertibility and marginal comparative statics.

The three major extensions are:

$$\text{Equilibrium:} \quad \max_R U(r(R), R), \tag{185}$$

$$\text{Asset Pricing:} \quad P_t = \mathbb{E}_t[M_{t+1} X_{t+1}], \tag{186}$$

$$\text{Econometrics:} \quad \mathbb{E}[\mathbf{m}_t(\boldsymbol{\theta})] = 0. \tag{187}$$

Together, these establish the proposed mathematical and research architecture.

24 Conclusion

This treatise has developed the Theory of Rational Return and Risk from its fundamental fractional-linear equation.

The first major result is the bilinear representation:

$$ArR + Br + CR + D = 0. \tag{188}$$

The second is the reciprocal Mobius representation:

$$r(R) = -\frac{CR + D}{AR + B}, \tag{189}$$

$$R(r) = -\frac{Br + D}{Ar + C}. \tag{190}$$

The third is the determinant condition:

$$AD - BC \neq 0. \tag{191}$$

The theory has subsequently been extended into investor optimization, portfolio equilibrium, rational asset pricing, stochastic discount factors, bond valuation, duration,

convexity, nonlinear econometrics, GMM, Bayesian estimation, dynamic systems and regime-dependent financial models.

Potential applications have also been identified across public finance, institutional economics, international relations, geopolitical risk, warfare economics, computational science and information systems.

The multidisciplinary extensions are research directions rather than claims that one financial equation universally governs all these fields.

The central scientific question remains:

$$\boxed{\begin{array}{c} Can\ financial\ return\ and\ risk\ be\ structurally\ represented\ by\ a \\ reciprocal\ fractional-linear\ transformation? \end{array}} \tag{192}$$

The answer must ultimately be established through theoretical derivation, econometric identification and reproducible empirical evidence.

The proposed theory therefore offers a mathematical structure, a financial-economic research programme and an econometric framework for investigating the relationship between rational return and risk.

A Algebraic Verification

Starting with

$$\frac{ar+b}{cr+d} = \frac{fR+g}{pR+q}, \tag{193}$$

cross-multiplication gives

$$aprR + aqr + bpR + bq = cfrR + cgr + dfR + dg. \tag{194}$$

Rearrangement gives

$$(ap-cf)rR + (aq-cg)r + (bp-df)R + (bq-dg) = 0. \tag{195}$$

Therefore:

$$A = ap - cf, \tag{196}$$
$$B = aq - cg, \tag{197}$$
$$C = bp - df, \tag{198}$$
$$D = bq - dg. \tag{199}$$

The determinant is

$$\Delta = AD - BC. \tag{200}$$

The first derivative is

$$\frac{dr}{dR} = \frac{\Delta}{(AR+B)^2}. \tag{201}$$

The second derivative is

$$\frac{d^2r}{dR^2} = -\frac{2A\Delta}{(AR+B)^3}. \tag{202}$$

B Dimensional Consistency

Suppose return is dimensionless and risk is measured in variance units.

Then the composite coefficients must satisfy:

$$[A] = [C], \tag{203}$$
$$[B] = [D], \tag{204}$$

where the brackets denote physical or measurement dimensions appropriate to the equation.

More explicitly, because r is dimensionless and R has units of squared return:

$$[ArR] = [CR], \tag{205}$$
$$[Br] = [D]. \tag{206}$$

Therefore:

$$[A] = [C], \tag{207}$$
$$[B] = [D]. \tag{208}$$

This dimensional analysis is necessary when risk definitions change.

C Numerical Implementation

A numerical implementation should impose:

$$AR_t + B \neq 0, \tag{209}$$
$$AD - BC \neq 0. \tag{210}$$

Optimization can use constrained nonlinear least squares or GMM.

The objective should include numerical penalties or admissible parameterizations to prevent optimization near singularities.

A complete implementation should record:

- Dataset version.
- Parameter normalization.
- Initial parameter values.
- Optimization algorithm.

- Convergence tolerance.
- Covariance estimator.
- Random seed, where applicable.
- Software version.
- Out-of-sample evaluation period.

D Suggested Computational Tests

1. Verify both algebraic inverse formulas.
2. Verify the determinant identity.
3. Verify the first and second derivatives.
4. Test parameter recovery using synthetic data.
5. Test behavior near the pole.
6. Test scale invariance.
7. Test recovery under measurement error.
8. Test estimation under endogeneity.
9. Test regime-dependent coefficients.
10. Compare forecasting performance against benchmark models.

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Glossary

APT Arbitrage Pricing Theory.

Bayesian Estimation Statistical inference combining prior distributions and observed data.

Conditional Variance Variance conditional on available information.

Convexity Second-order sensitivity of a financial price to changes in a specified variable.

Endogeneity Correlation between an explanatory variable and the structural disturbance.

Expected Shortfall Expected loss conditional on exceeding a specified loss quantile.

GARCH Generalized Autoregressive Conditional Heteroskedasticity.

GMM Generalized Method of Moments.

Instrumental Variable A variable satisfying relevance and exogeneity conditions for identifying endogenous relationships.

Mobius Transformation A fractional-linear transformation of the form $(ax+b)/(cx+d)$.

Nonlinear Least Squares An estimation procedure minimizing squared nonlinear residuals.

Risk Premium Expected return in excess of the risk-free return.

Risk-Free Rate The return associated with the specified risk-free benchmark.

Stochastic Discount Factor A random variable used to price future financial payoffs.

Structural Break A change in the parameters of an econometric relationship.

Systemic Risk Risk of disruption to the financial system as a whole.

Welfare Economics The study of economic efficiency, distribution and social welfare.

The End